Database Technologies – Assignment 10 PL-SQL

1. Write a procedure that computes the perimeter and the area of a rectangle. Define your own values for the length and width. (Assuming that L and W are the length and width of the rectangle, Perimeter = 2*(L+W) and Area = L*W.

```
KD3-Prerana-96192>DELIMITER //
KD3-Prerana-96192>CREATE PROCEDURE sp_calculate(len INT, width INT)
    -> BEGIN
    ->      DECLARE perimeter INT;
    ->      DECLARE area INT;
    ->      SELECT 2*(len + width ) INTO perimeter;
    ->      SELECT len*width INTO area;
    ->      SELECT area, perimeter ;
    -> END ;
    -> //
Query OK, 0 rows affected (0.03 sec)

KD3-Prerana-96192>DELIMITER ;
KD3-Prerana-96192>CALL sp_calculate(6, 4);
+------+-----------+
| area | perimeter |
+------+-----------+
|   24 |        20 |
+------+-----------+
1 row in set (0.01 sec)

Query OK, 0 rows affected (0.01 sec)
```

2. Write a procedure that declares an integer variable called num, assigns a value to it, computes and inserts into the temp table the value of the variable itself, its square, and its cube.

```
KD3-Prerana-96192>CREATE TABLE result(num INT, squares INT, cubical INT);
Query OK, 0 rows affected (0.04 sec)
```

```
KD3-Prerana-96192>DELIMITER //
KD3-Prerana-96192>CREATE PROCEDURE sp_var(num INT)
    -> BEGIN
    ->     DECLARE squares INT;
    ->     DECLARE cubical INT;
    ->     SELECT num*num INTO squares;
    ->     SELECT num*num*num INTO cubical;
    ->     INSERT INTO result VALUES(num, squares, cubical);
    -> end;
    -> //
Query OK, 0 rows affected (0.02 sec)

KD3-Prerana-96192>DELIMITER ;
KD3-Prerana-96192>CALL sp_var(6);
Query OK, 1 row affected (0.01 sec)

KD3-Prerana-96192>SELECT * FROM  result;
+------+---------+---------+
| num  | squares | cubical |
+------+---------+---------+
|    6 |      36 |     216 |
+------+---------+---------+
1 row in set (0.00 sec)
```

3. Create a procedure to Convert a temperature in Fahrenheit (F) to its equivalent in Celsius (C) and vice versa. The required formulae are C= (F-32) *5/9 F= 9/5*C + 32

```
KD3-Prerana-96192>CREATE PROCEDURE sp_temperature1(val CHAR, num FLOAT)
    -> BEGIN
    ->      DECLARE degree_Cel FLOAT;
    ->      DECLARE degree_Far FLOAT;
    ->
    ->      if (val = 'C')
    ->      THEN
    ->      SET degree_Far = (9/5)* num + 32;
    ->      end if;
    ->
    ->      if (val = 'F')
    ->      THEN
    ->      SET degree_Cel = ((num-32)*5)/9;
    ->      end if;
    ->
    ->
    ->     SELECT degree_Cel, degree_Far;
    -> END;
    -> //
Query OK, 0 rows affected (0.01 sec)

KD3-Prerana-96192>DELIMITER ;
KD3-Prerana-96192>CALL sp_temperature1('C', 32);
+------------+------------+
| degree_Cel | degree_Far |
+------------+------------+
|       NULL |       89.6 |
+------------+------------+
1 row in set (0.00 sec)

Query OK, 0 rows affected (0.00 sec)
```

4. Create a procedure to Convert a number of inches into yards, feet, and inches. For example, 124 inches equals 3 yards, 1 foot, and 4 inches.

```
KD3-Prerana-96192>DELIMITER //
KD3-Prerana-96192>CREATE PROCEDURE sp_measure(num FLOAT )
    -> BEGIN
    ->      DECLARE inches FLOAT;
    ->      DECLARE yards FLOAT;
    ->      DECLARE feets FLOAT;
    ->      SELECT num INTO inches;
    ->      SELECT num/12 INTO feets;
    ->      SELECT num/36 INTO yards;
    ->      SELECT concat(inches,'inches equals, ', feets,' feet, and', yards,' yards')  as val
    -> END ;
    -> //
Query OK, 0 rows affected (0.01 sec)

KD3-Prerana-96192>DELIMITER ;
KD3-Prerana-96192>CALL sp_measure( 216);
+--------------------------------------+
| val                                  |
+--------------------------------------+
| 216inches equals, 18 feet, and6 yards |
+--------------------------------------+
1 row in set (0.00 sec)

Query OK, 0 rows affected (0.00 sec)
```

5. Create a procedure to read two real numbers and tell whether the product of the two numbers is equal to or greater than 100.

```
KD3-Prerana-96192>DELIMITER //
KD3-Prerana-96192>
KD3-Prerana-96192>CREATE PROCEDURE sp_cpr(num INT, num1 INT)
    -> BEGIN
    ->      DECLARE mul INT;
    ->      SET mul = num * num1;
    ->
    ->      if (mul >100) THEN
    ->         SELECT  CONCAT(mul,' is greater than 100 ');
    ->      else
    ->         SELECT  CONCAT('The value is ',mul);
    ->      end if;
    ->
    -> END;
    -> //
Query OK, 0 rows affected (0.01 sec)

KD3-Prerana-96192>DELIMITER ;
KD3-Prerana-96192>
KD3-Prerana-96192>CALL sp_cpr( 12,12);
+-------------------------------------+
| CONCAT(mul,' is greater than 100 ') |
+-------------------------------------+
| 144 is greater than 100             |
+-------------------------------------+
1 row in set (0.00 sec)

Query OK, 0 rows affected (0.00 sec)
```

```
KD3-Prerana-96192> drop procedure sp_cpr;
Query OK, 0 rows affected (0.01 sec)

KD3-Prerana-96192>DELIMITER //
KD3-Prerana-96192>
KD3-Prerana-96192>CREATE PROCEDURE sp_cpr(num INT, num1 INT)
    -> BEGIN
    ->      DECLARE mul INT;
    ->      SET mul = num * num1;
    ->
    ->      if (mul >100) THEN
    ->         SELECT  CONCAT(mul,' is greater than 100 ');
    ->      else
    ->         SELECT  CONCAT('The value is ',mul);
    ->      end if;
    ->
    -> END;
    -> //
Query OK, 0 rows affected (0.01 sec)

KD3-Prerana-96192>DELIMITER ;
KD3-Prerana-96192>
KD3-Prerana-96192>CALL sp_cpr( 0,12);
+----------------------------+
| CONCAT('The value is ',mul) |
+----------------------------+
| The value is 0             |
+----------------------------+
1 row in set (0.00 sec)
```